Value & Publishability

Rubric Deliverable P4 — MRI-ReportGenerator (Cervical Spine MRI Analysis)

EECE503N Final Project

2026-06-08

Framing. The project is positioned as an **Application** (rubric GT5). Therefore the bar this deliverable must clear is *application value*; *publishability* is presented as a credible *upside*, not as a claim. We do not assert a guaranteed publication — the rubric's Research track carries a publishability hard-stop, and over-claiming would be dishonest. What we do show is (i) concrete, defensible application value, and (ii) specific, benchmarkable research nuggets with named candidate venues. Companion: `deliverables/T1_ai_depth.tex` (technical depth), `deliverables/P2_baseline.tex` (the quantified non-AI baseline), and the full `paper/`.

1 Application value (the bar)

The system augments — never replaces — the radiologist read of a cervical sagittal-T2 MRI. Its value is in four properties the manual baseline lacks:

Property	What the system delivers	Why it matters vs. the manual baseline
Reproducibility	Deterministic geometric measurements — the same study yields the same numbers every run	Manual measurement has real inter-reader (and intra-reader) variability; the quantified gap is in P2
Speed	Measurements computed in seconds per case (per-component latency measured; see infra `tradeoffs.md`)	Manual caliper measurement of canal, disc, Cobb and vertebral geometry is slow; the per-case time is in P2
Auditability	Every flag traces to a measurement and a *cited* threshold (the G6 catalog), not an opaque verdict	A radiologist can check each number; a black-box score cannot be audited or defended
Triage / augmentation	Pre-populates the structured measurement table and flags cases whose measurements cross reference thresholds for review	Frees specialist time from repetitive geometry for judgement; "flagged for physician review," never a diagnosis

Who benefits / who pays. Radiology departments and research-hospital workflows: the buyer is the imaging department or research group that wants consistent, fast, auditable cervical measurements at volume. The value proposition is *consistency + speed + auditability*, not a black-box diagnosis.

The value is quantified by the baseline contrast (P2). The two numbers that make this case concrete — manual measurement/reading *time* per case, and manual inter-observer *variability* — are the subject of deliverable P2. The argument here is structural: a deterministic pipeline removes the variability entirely and collapses the time, with every output cited. P2 supplies the magnitudes.

2 Publishability (the upside)

Independent of the application framing, the project produced research-grade artifacts that are genuinely benchmarkable. We list them honestly, separating what is a real contribution from what is a heuristic.

2.1 What is novel / benchmarkable

- **Healthy-anchored, disease-agnostic geometric detectors.** Measurements are validated to read *inside* a healthy range and to be crossed by symptomatic anatomy, rather than trained on a disease label. This is a reusable evaluation pattern for measurement pipelines where per-case ground truth is unavailable.
- **Threshold-crossing validation under a no-ground-truth constraint.** A reproducible methodology (healthy anchor + distribution separation by Mann–Whitney) that yields a defensible validation *without* per-case radiologist labels — because no public dataset pairs cervical MRI with per-case expert measurements. The methodology itself is the contribution.
- **The physical-dimension vs. intensity scanner-immunity finding.** We show empirically that millimetre metrics (canal, SAC, cord) separate cross-dataset while intensity/ratio metrics (disc signal, height ratios) are acquisition-confounded and must be validated within a single dataset. This is a non-obvious, transferable insight for anyone validating MRI-derived measurements across scanners.
- **Honest negative results.** The disc *signal* axis does *not* discriminate (AUC 0.50, level-controlled), and disc AP-width's apparent strong effect collapsed under level-stratification (AUC 0.79 $\rightarrow$ 0.61). Reported negatives and confound corrections are publishable precisely because they are rare in this literature.
- **A reproducible validation on real healthy + symptomatic cervical MRI** (Spine-Generic healthy anchor; MMCSD symptomatic demonstration set), with a committed harness and a fully-cited threshold catalog.

2.2 What is ***not*** claimed

The per-disc and per-vertebra cut-points (e.g. the disc/VB ratio threshold) have no κ/AUC ground truth, so they are scoped as *research-grade heuristics*, not validated clinical thresholds. The ceiling on the disc metrics is data-limited (a coarse binary lesion label, and features the segmentation masks cannot see); raising it requires radiologist per-disc grades, which is future work, not a present claim.

2.3 Candidate venues

The work fits the medical-imaging-informatics / computational-spine niche rather than a new-architecture venue. Honest candidates, in order of fit:

- **MICCAI workshops** — e.g. Computational Spine Imaging (CSI) or applications/validation-focused workshops: the methodology + reproducible validation is a natural fit.
- **SPIE Medical Imaging** (Computer-Aided Diagnosis / Image Processing tracks) — measurement pipelines with validation are in scope.

- **Journal of Digital Imaging / Insights into Imaging** — clinical imaging informatics; a deployed, auditable measurement-and-report system with a validation study is on-topic.

The `paper/` draft (18 sections + appendices, Overleaf-ready) is the artifact that would be adapted to a chosen venue's format.

3 Summary for the grader

- **Application value is real and structural**: reproducibility, speed, auditability, and triage — with the quantified baseline contrast supplied by P2.
- **Publishability is a credible upside, not a claim**: the validation methodology, the scanner-immunity finding, and the honest negative results are genuinely benchmarkable, with named candidate venues and an existing paper draft.
- **We are explicit about the boundary**: heuristic thresholds are labelled as such; the data-limited ceiling is stated; no guaranteed publication is asserted.